

From four t(w)o three: An exploration of interferometric 4D-STEM for measuring 3D variations in twisted bilayer graphene

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Four-dimensional (4D) scanning transmission electron microscopy (STEM) has become an almost routine procedure in high-level electron microscopy due to the combination of modern pixelated direct-electron detectors and computers able to handle the massive resulting diffraction data. However, computational power is merely the muscle, and requires solid theory and steadfast thought behind the employed analysis. Therefore, a lot of effort has gone into developing ways to coax novel information from 4D-STEM datasets. While some methods, like ptychography or center of mass imaging, are by now well established, others remain challenging. Here, we discuss interferometric 4D-STEM, a method still in its adolescence, that could shed light on a property well hidden from conventional STEM, the three-dimensional (3D) structure of thin specimen, in particular twisted bilayer graphene. There, the overlap of the identical but twisted layers leads to the formation of periodic moiré patterns, superstructures that cause the periodic modulation of various properties such as the distance between the layers, which varies depending on the local stacking order [1]. Interference patterns formed in the overlap regions of convergent-beam diffraction disks originating from the separate layers when probed with a defocused beam contain information on the subtle interlayer distance variations. Careful analysis could thus provide a spatially resolved distance map over the whole moiré pattern. Building upon the works by Latychevskaia [2] and Zachman [3], who established the foundation of this method, we examine whether, to what extent, with which resolution, and under what conditions interferometric 4D-STEM yields sensible results, by systematically testing it against simulated data of increasing complexity. This systematic study aims to lay the groundwork for future experimental work and provide a useful starting point for people interested in doing similar work. With the nature of data and also of science itself changing in the wake of technological advances, it is more important than ever to share our work and support one another in our quest for good, reliable, and open research.

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